Flume Notes: 2025.03.27

- GCrew site called "Flume"
- Need a name for project
- Put time w/ time zone (i.e EDT)
- Need large bucket to get samples so everything comes from 1 source
- Bottle caps for DIC (GC + TOC samples)
- Labels w/ Room for Date, Time, Site, Type Sample
- Leave room on Alk sample for xCO_2 & Salinity

YSI Values for flume sample: DO % = 127.2 SpC/C = 12268/9341
set temp, xCO_2 from DO mg/l = 13.47 Sal = 7.16
flume data when downloaded

Flow Rate @ ~ 0.5 LPM

- Downloaded RAD7 - flume data

DOCK NOTES:

LICOR: CH_4 = 11950 ppb CO_2 160 ppm * NOT ON logger currently

CO_2 Scout: 12.4 CO_2 = 107 ppm P = 1020

- Downloaded RAD7 - DOCK Data

DO % = 121 SpC/C = 17324/12729 > BUCKET
DO mg/L = 12.28 Sal = 10.17

- FLOW RATE @

DO % = 129.6 SpC/C = 17430/12557 > Actual Water
DO mg/L = 13.67 Sal = 10.29

April 3, 2025 Sampling Dock, Flume

protocol notes:

- Have 2.5-3 gallon bucket to use for all samples. Rinse 3x then gently fill to brim → time of sample, note CO_2 / CH_4
- Before sampling turn on YSI probe ~10 min then calibrate for DO %.
- Take YSI meas lst, then DIC, sulfates, sulfides then rest of parameters

Dock 3 April 2025 @ 13:00

L-COR CO_2 = 154 ppm Scout = ATM meas. (123 previous)

CH_4 = 15133 ppb T = 15.6

YSI bucket: Sal = 10.19

DO % = 104.1

DO mg/L = 10.78

SpC / C = 14772 / 11822

T = 14.2

Flume April 3, 2025 @ 14:43

Scout [xCO_2 = 2100, T = 15.4]

YSI:

DO % = 91.4

mg/L = 8.40

SpC / C = 16331 / 13526

Sal = 9.61

T = 16

Flume - low tide
air: 80°C breezy

14:00 EDT

27.3 °C
51.4 DO%
3.~~83~~ DO mg/L
15407 spC uS/cm
15569 C uS/cm
9.63 Sal ppt
7.33 pH
-57.3 pH mV

TA ✓
DOC ✓
DICex ✓
DIC ✓✓
NUTR ✓
SO_4/H_2S ✓

Set up RAD7 $V'd system

~~00:00~~

CK

15:00 EDT

21.7 °C
109.4 DO%
9.15 DO mg/L
17465 spC uS/cm
16368 C uS/cm
10.31 Sal
8.98 pH
-147.7 pH

TA ✓
DOC ✓
DICex ✓
DIC ✓✓
NUTR ✓
SO_4/H_2S ✓

Set up RAD
$V'd SYS.

COMPASS Synoptic Aquatroll calibration sheet

Date (s)	
Weather	
People	

Maintenance schedule

Calibrate	Every 3 months
pH sol./junction	As needed
Change DO cap	2 years / if damaged

Troll Protocol

Site	Location	Time*	Sensor SN	DO (%)**	Cond (uS/cm)**	pH 4**	pH 7**	pH 10**	Dirty/pre/post	Check SDI

*time sensor is removed from well; **record pre and post-cal values; ***time well is fully pumped

Notes: